

Transformations & Constructions

Answer sheet · 27 marks total

Section A — Translation

A1. $(2 + 3, 5 - 4)$ (1) = **(5, 1)** (1)

A2. $4 - 1 = 3$ (right), $-2 - 1 = -3$ (down) (1) vector = **(3, -3)** (1)

A3. **False** (1)

Section B — Rotation

B1. **Angle** of rotation (1) **direction** — clockwise or anticlockwise (1) **centre** of rotation (1)

B2. $(-2, -3)$ (1) correct use of the 180° rule $(x,y) \rightarrow (-x,-y)$ (1)

Section C — Reflection

C1. **(-3, 2)** (1)

C2. **(4, 1)** (1)

C3. Distance from $x=2$ to $x=5$ is 3, image $x = 2-3 = -1$ (1) image = **(-1, 1)** (1)

Section D — Enlargement

D1. $4 \times 3 =$ **12 cm** (1)

D2. The shape becomes **smaller** (half the size) (1)

D3. Area $SF = 3^2 = 9$ (1) $8 \times 9 =$ **72 cm²** (1)

D4. Image is **twice the size** of the object (1) image is on the **opposite side** of the centre, turned 180° (1)

D5. The image is still formed by the same ray-from-centre process (just with $SF < 1$), so it is still classed as an enlargement even though it gets smaller (1)

Section E — Constructions

E1. Set compasses to a radius more than half of AB (1) draw arcs from A and from B (same radius) so they cross above and below the line (1) join the two crossing points with a straight line (1)

E2. Draw an arc from the vertex crossing both arms of the angle (1) from each of those crossing points draw equal-radius arcs that meet inside the angle, then join the vertex to that meeting point (1)

E3. They show the method used and earn marks — an examiner cannot award construction marks for a "correct-looking" line with no working shown (1)

Marking note: award method marks for a correctly set-up calculation or equation even if the final answer contains an arithmetic slip. For construction questions, accept any correctly worded description of the method (exact wording does not need to match). Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.